

Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behavior

**Project Report Submitted in fulfillment of the
Requirements for the Award of the Internship of
Summer Training Program by IDRA**

Data Science and AI

By:

Name: Akash Barman

Institute: University of Calcutta

Institute Roll No.: T91/CSE/236001

Enrollment No.: IDRA-2026-444610

Under the Supervision of

Dr. Shaheena Salam

Through the platform:



India Data Research Academy

DECLARATION

I hereby declare that the project report titled “**Predicting Customer Churn: A Data-Driven Analysis of Subscription Customer Behaviour**” is my own work completed as part of the IDRA Data Science and AI Capstone Project. The data analysis, preprocessing, statistical analysis, feature engineering, machine learning implementation, and interpretation presented in this report were carried out by me. All external datasets, references, and sources used in this project have been appropriately acknowledged.

ACKNOWLEDGEMENT

I sincerely express my gratitude to the **India Data Research Academy (IDRA)** for providing the training and capstone project opportunity in Data Science and AI. I am grateful to **Dr. Shaheena Salam**, my project supervisor, for her guidance and support throughout the project. I also acknowledge the source of the **Telco Customer Churn dataset** used for this analysis. Finally, I would like to thank my institute, instructors, and everyone who provided support and encouragement during the completion of this project.

ABSTRACT

Customer churn is an important business problem for subscription-based service providers because identifying customers who are likely to discontinue their services can support data-driven customer retention strategies. This project analyses customer subscription data to identify patterns associated with customer churn and develops machine learning models for churn classification.

The analysis uses the Telco Customer Churn dataset. After data cleaning, the dataset contained 7,032 customer records. Exploratory data analysis was performed to examine relationships between churn and customer characteristics such as tenure, monthly charges, contract type, and internet service type. Statistical analysis was conducted using the Mann–Whitney U test and chi-square test to examine differences and associations between churn status and selected variables. Feature engineering was also performed by creating an average monthly spending feature.

Two classification models, Logistic Regression and Decision Tree, were developed and evaluated using accuracy, precision, recall, and F1-score. On the testing dataset, Logistic Regression achieved 80.38% accuracy, 64.85% precision, 57.22% recall, and 60.80% F1-score, while the Decision Tree achieved 78.39% accuracy, 58.62% precision, 63.64% recall, and 61.03% F1-score.

The findings demonstrate that customer churn can be investigated effectively through exploratory analysis, statistical testing, feature engineering, and machine learning.

Keywords: Customer Churn; Data Science; Machine Learning; Classification; Customer Behaviour

TABLE OF CONTENTS

1. Introduction
2. Problem Statement and Motivation
3. Objectives
4. Research Questions
5. Dataset Description and Data Understanding
6. Data Cleaning
7. Data Preprocessing
8. Exploratory Data Analysis
9. Statistical Analysis
10. Feature Engineering
11. Model Development
12. Model Evaluation
 - 12.1 Classification Evaluation
 - 12.2 Overfitting and Underfitting
13. Results and Findings
14. Discussion
15. Limitations
16. Conclusion
17. Recommendations and Future Work
 - 17.1 Recommendations
 - 17.2 Future Work
18. References

Appendix A. Code

- A.1 Data Loading and Inspection
- A.2 Data Cleaning
- A.3 Preprocessing
- A.4 EDA and Visualization
- A.5 Model Training and Evaluation

Appendix B. Additional Figures and Tables

- B.1 Additional Model Evaluation Results
- B.2 Training and Testing Accuracy

1. INTRODUCTION

1.1 Background

Customer churn refers to the situation in which a customer stops using a company's services. For subscription-based service providers, understanding customer churn is important because customer behaviour can vary according to factors such as service usage, contract type, tenure, and monthly charges. Data analysis and machine learning can be used to identify patterns associated with churn and to support the systematic analysis of customers who may be more likely to leave.

1.2 Context of the Problem

Subscription service providers need to understand customer behaviour in order to analyse patterns related to customer retention and churn. The stakeholders in this context include service providers, customer relationship teams, and business analysts who can use customer data to examine characteristics associated with churn. A data-driven analysis can help identify relationships between customer attributes and churn status and provide evidence that can support customer retention planning.

1.3 Scope of the Study

This study focuses on analysing customer subscription data and predicting the churn status of customers using machine learning classification techniques. The study covers data cleaning, preprocessing, exploratory data analysis, statistical analysis, feature engineering, model development, and model evaluation. The analysis is limited to the variables available in the selected dataset and does not establish causal relationships between customer characteristics and churn.

2. PROBLEM STATEMENT AND MOTIVATION

The problem addressed in this project is to determine whether customer subscription characteristics can be used to identify patterns associated with customer churn and to develop machine learning models capable of classifying customers according to their churn status.

The motivation for this study is to apply data science techniques to a practical customer behaviour problem. Analysing churn patterns can provide a structured understanding of how variables such as contract type, tenure, monthly charges, and internet service relate to churn status.

3. OBJECTIVES

- To understand and prepare the customer subscription dataset for analysis.
- To identify important patterns and relationships between customer characteristics and churn status.
- To examine differences and associations between churned and non-churned customers using statistical analysis.
- To develop machine learning classification models for predicting customer churn.
- To evaluate the developed models using accuracy, precision, recall, F1-score, and confusion matrices.

4. RESEARCH QUESTIONS

1. What are the major characteristics and data quality issues present in the customer subscription dataset?
2. Which customer characteristics show meaningful relationships with churn status?
3. How do customer tenure, monthly charges, contract type, and internet service vary according to churn status?

4. Are the observed differences and associations between selected customer characteristics and churn statistically significant?
5. How accurately can machine learning models classify customers according to their churn status?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

The dataset used in this project is the Telco Customer Churn dataset, which contains customer-level information related to telecommunications services and customer churn status. The dataset was provided for the capstone project and was used for the purpose of analyzing customer behavior and developing churn classification models.

5.2 Dataset Size and Structure

Attribute	Description	Value	Role
Number of Records	Total customer records	7,043	Dataset Size
Number of Variables	Total columns	21	Dataset Structure
Target Variable	Customer churn status	Churn	Target
Identifier Variable	Unique customer identifier	customerID	Identifier
Final record	Records remaining after cleaning	7,032	Cleaned Dataset

Table 1. Summary of the Dataset

5.3 Data Dictionary

Variable	Meaning	Data Type	Role
customerID	Unique customer identifier	Object	Identifier
gender	Customer gender category	Object	Feature
SeniorCitizen	Indicates senior-citizen status	Integer	Feature
tenure	Number of months with the service	Integer	Feature
Contract	Customer contract type	Object	Feature
InternetService	Type of internet service	Object	Feature
MonthlyCharges	Monthly amount charged	Float	Feature
TotalCharges	Total amount charged	Float	Feature
PaymentMethod	Customer payment method	Object	Feature
Churn	Customer churn status	Object	Target

Table 2. Data Dictionary

6. DATA CLEANING

The raw dataset was examined for missing values, duplicate records, inconsistent categorical values, incorrect data types, and potential outliers before modelling. The cleaning process was performed systematically to preserve valid customer records while correcting data-quality issues that could affect subsequent analysis.

6.1 Missing Values

Initially, no missing values were detected by the standard missing-value check. However, during conversion of the `TotalCharges` variable from text to numeric format, **11 blank/invalid entries** were identified. These records corresponded to customers with zero tenure and were removed from the modelling dataset. After this treatment, the final dataset contained **no missing values**.

6.2 Duplicate Records

The dataset contained **0 duplicate rows**. Therefore, no records were removed due to duplication.

6.3 Data Types and Inconsistent Values

The **TotalCharges** variable was initially stored as an object/text data type even though it represented numerical billing amounts. It was converted to a numeric floating-point format using appropriate numeric conversion. Categorical variables were also inspected for inconsistent values, and no major category inconsistencies were identified.

6.4 Outlier Handling

Potential outliers in the numerical variables **tenure**, **MonthlyCharges**, and **TotalCharges** were examined using the Interquartile Range (IQR) method. The lower and upper bounds were calculated using $1.5 \times \text{IQR}$ from the first and third quartiles. The analysis identified zero outliers in all three variables. Therefore, no observations were removed, capped, or transformed specifically for outlier treatment. This approach preserved the original valid customer records while confirming that no extreme values required additional handling.

6.5 Before vs After Cleaning

Data Quality Aspect	Before Cleaning	Cleaning Action	After Cleaning
Records	7,043	Removed 11 invalid <code>TotalCharges</code> records	7,032
Missing Values	11 invalid/blank <code>TotalCharges</code> entries identified after numeric conversion	Removed affected records	0
Duplicate Records	0	No action required	0
Data Type Issue	<code>TotalCharges</code> stored as object	Converted to numeric	<code>float64</code>
Categorical Inconsistencies	No major inconsistencies identified	No action required	Consistent
Numerical Outliers	0 detected by IQR method	No action required	0

Table 3. Data Quality Before and After Cleaning

7. DATA PREPROCESSING

After data cleaning, the dataset was prepared for machine learning through feature engineering, categorical encoding, numerical scaling, and train-test splitting. The target variable **Churn** was separated from the predictor variables. The preprocessing steps were implemented using a **ColumnTransformer** so that numerical and categorical variables could be handled appropriately.

7.1 Categorical Encoding

The dataset contains several categorical variables such as gender, Partner, Dependents, Contract, InternetService, and PaymentMethod. Since machine learning models require numerical input, categorical variables were converted into numerical representations using One-Hot Encoding. The **OneHotEncoder** was configured with `handle_unknown="ignore"` to ensure that previously unseen categories would not cause errors during prediction.

7.2 Feature Scaling and Transformation

Numerical variables were standardized using **StandardScaler**. The numerical variables included **SeniorCitizen**, **tenure**, **MonthlyCharges**, **TotalCharges**, and the engineered feature **AvgMonthlySpend**. Standardization was applied to place numerical variables on a comparable scale. The preprocessing transformation was fitted using only the training data and subsequently applied to the test data to avoid data leakage.

7.3 Train-Test Split

The cleaned dataset was divided into training and testing subsets using an 80:20 split. The **Churn** variable was used as the target variable, while the remaining variables were used as predictors. Stratified sampling was applied to preserve the proportion of churned and non-churned customers in both subsets. A random state of 42 was used to ensure reproducibility.

The resulting training dataset contained **5,625 records**, while the testing dataset contained **1,407 records**.

7.4 Final Feature Set

After removing the **customerID** identifier and separating the target variable, 20 predictor features were available for modelling, including the engineered **AvgMonthlySpend** feature. The preprocessing pipeline transformed these variables into **46 numerical features** through one-hot encoding and numerical scaling. The same fitted preprocessing procedure was applied consistently to both training and testing data.

8. EXPLORATORY DATA ANALYSIS

8.1 Univariate Analysis

```
# Churn distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df_model, x="Churn")

plt.title("Customer Churn Distribution")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Number of Customers")

plt.show()
```

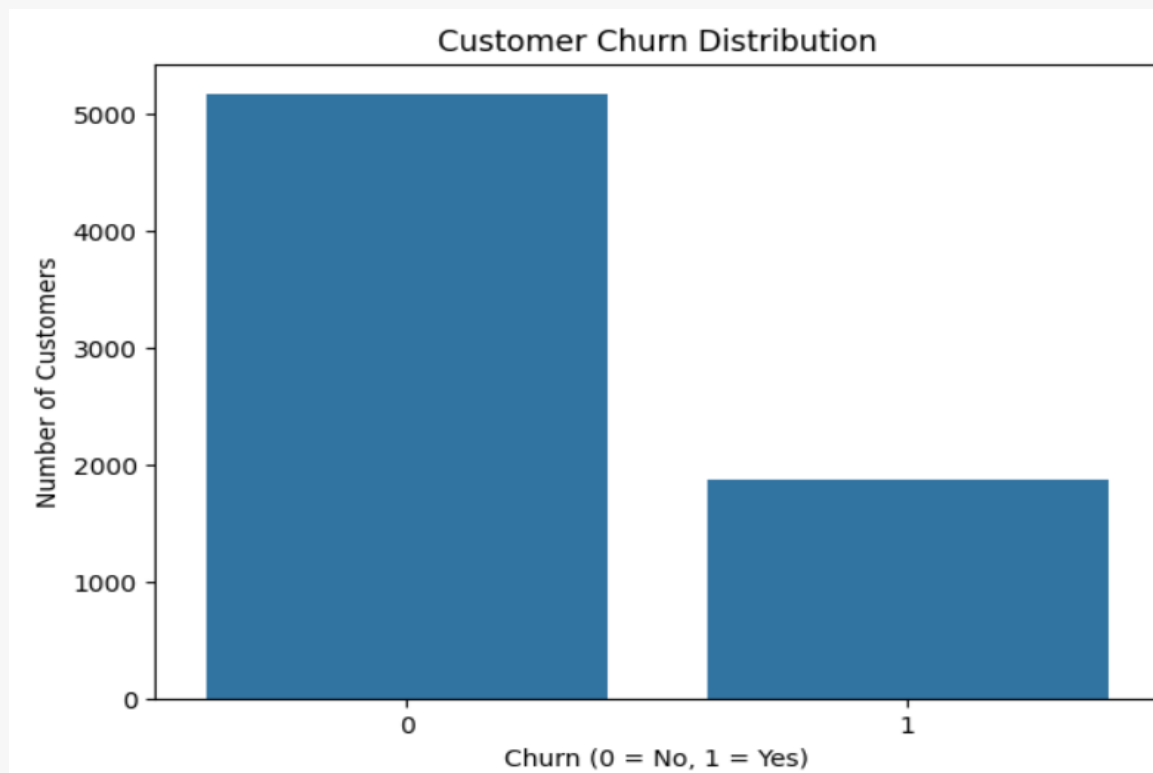


Figure 1. Distribution of Customer Churn

The distribution of the target variable was examined to understand the overall churn pattern in the dataset. Among the 7,032 cleaned customer records, 5,163 customers did not churn, while 1,869 customers churned. Thus, the overall churn rate was **26.58%**. The target distribution indicates that the dataset contains more non-churned than churned customers.

8.2 Categorical Analysis

```
# Contract type vs customer churn

plt.figure(figsize=(8, 5))

sns.countplot(data=df_model, x="Contract", hue="Churn")

plt.title("Customer Churn by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
plt.legend(title="Churn", labels=["No", "Yes"])

plt.xticks(rotation=10)
plt.show()
```

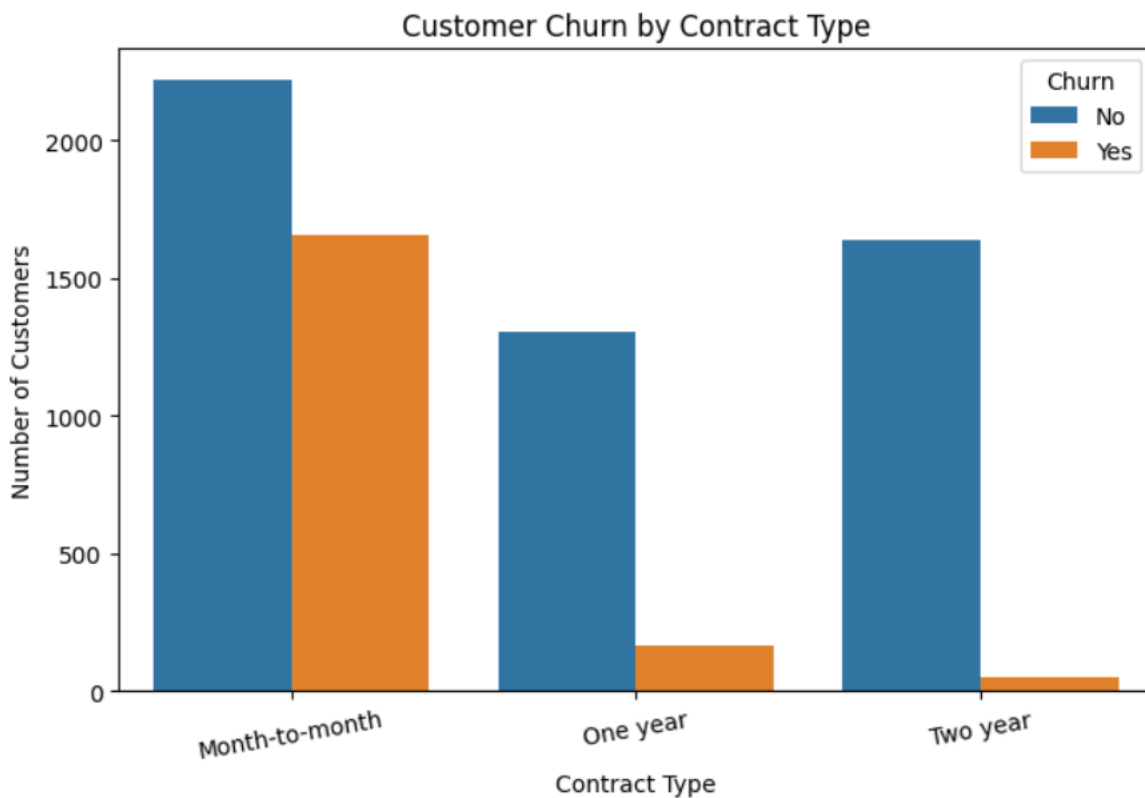


Figure 2. Customer Churn Across Contract Types

Contract type was examined to compare churn patterns across customer groups. The distribution shows substantial differences in churn between contract categories. Customers with month-to-month contracts account for a considerably larger share of churned customers compared with customers under longer-term contracts. This categorical pattern provides an important basis for further statistical analysis of the relationship between contract type and churn.

8.3 Bivariate / Relationship Analysis

```
# Monthly charges by churn status

plt.figure(figsize=(8, 5))

sns.boxplot(data=df_model, x="Churn", y="MonthlyCharges")

plt.title("Monthly Charges by Churn Status")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Monthly Charges")
plt.xticks([0, 1], ["No Churn", "Churn"])

plt.show()
```

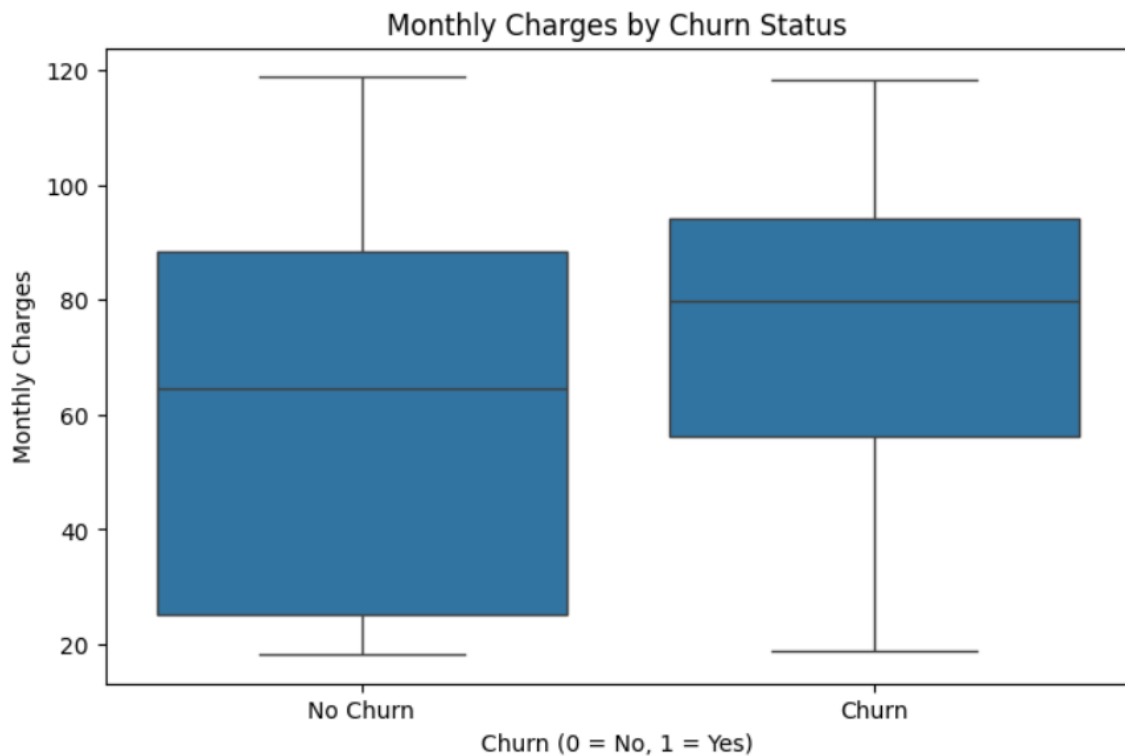


Figure 3. Relationship Between Monthly Charges and Customer Churn

The relationship between monthly charges and customer churn was examined by comparing the distribution of MonthlyCharges across the two churn groups. Churned customers had higher monthly charges on average than non-churned customers, with mean monthly charges of 74.44 and 61.31 respectively. This indicates an observable association between monthly charges and churn status in the dataset. However, this relationship should not be interpreted as evidence that higher charges directly cause customers to churn.

8.4 Correlation Analysis

```
# ===== CORRELATION ANALYSIS =====

correlation_data = df[[
    "SeniorCitizen",
    "tenure",
    "MonthlyCharges",
    "TotalCharges"
]]
correlation_matrix = correlation_data.corr()

plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Variables")
plt.tight_layout()
plt.show()
```

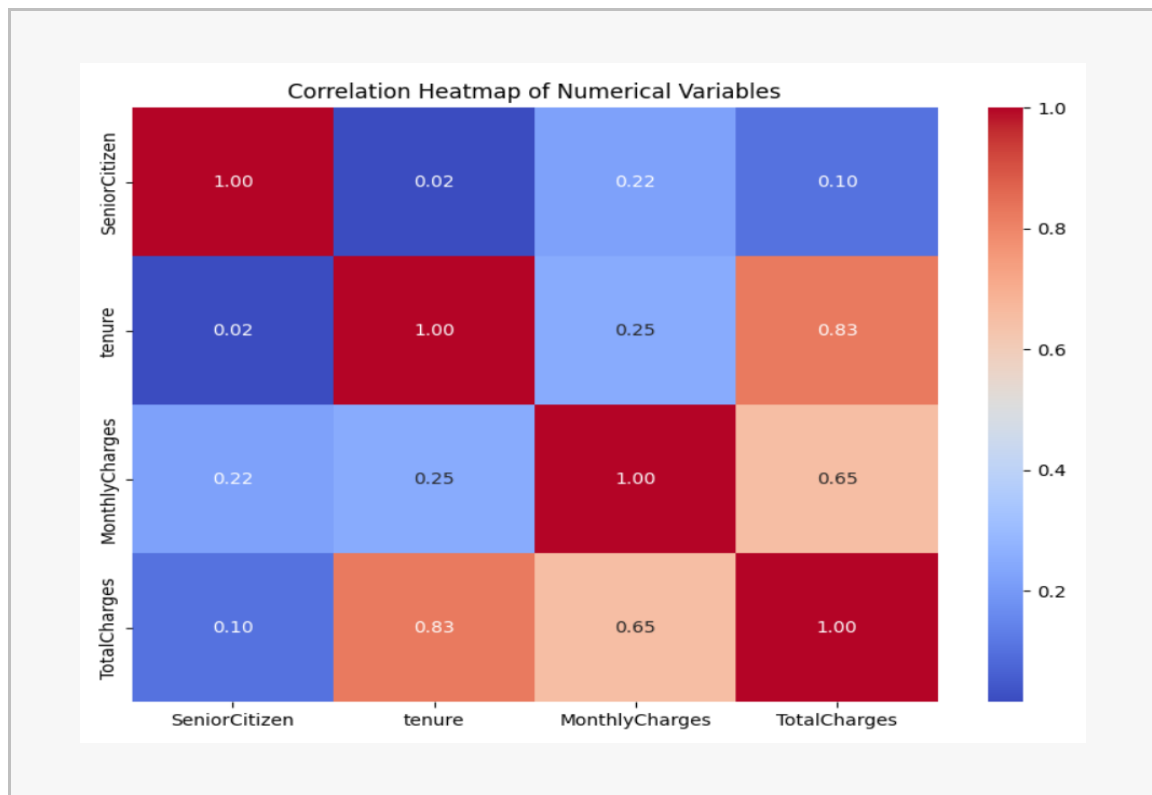


Figure 4. Correlation Heatmap of Numerical Variables

A correlation analysis was performed on the numerical variables using Pearson correlation coefficients. The strongest positive relationship was observed between `tenure` and `TotalCharges`, with a correlation coefficient of **0.83**. This indicates a strong positive linear association, meaning customers with longer service tenure generally tend to have higher total charges. `MonthlyCharges` also showed a positive relationship with `TotalCharges` ($r = 0.65$), while the relationship between `tenure` and `MonthlyCharges` was relatively weak ($r = 0.25$). `SeniorCitizen` showed weak correlations with the other numerical variables. These relationships were considered when interpreting the dataset and preparing the features for modelling. Correlation indicates association and does not establish causation.

8.5 Other Figures

Figure 5 — Tenure vs Churn

The tenure distribution shows that churned customers generally have shorter service tenure than customers who did not churn. The median tenure of churned customers was substantially lower, indicating that churn was more concentrated among customers with shorter service histories.

```
# Tenure distribution by churn status

plt.figure(figsize=(8, 5))

sns.boxplot(data=df_model, x="Churn", y="tenure")

plt.title("Customer Tenure by Churn Status")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Tenure (Months)")
plt.xticks([0, 1], ["No Churn", "Churn"])

plt.show()
```

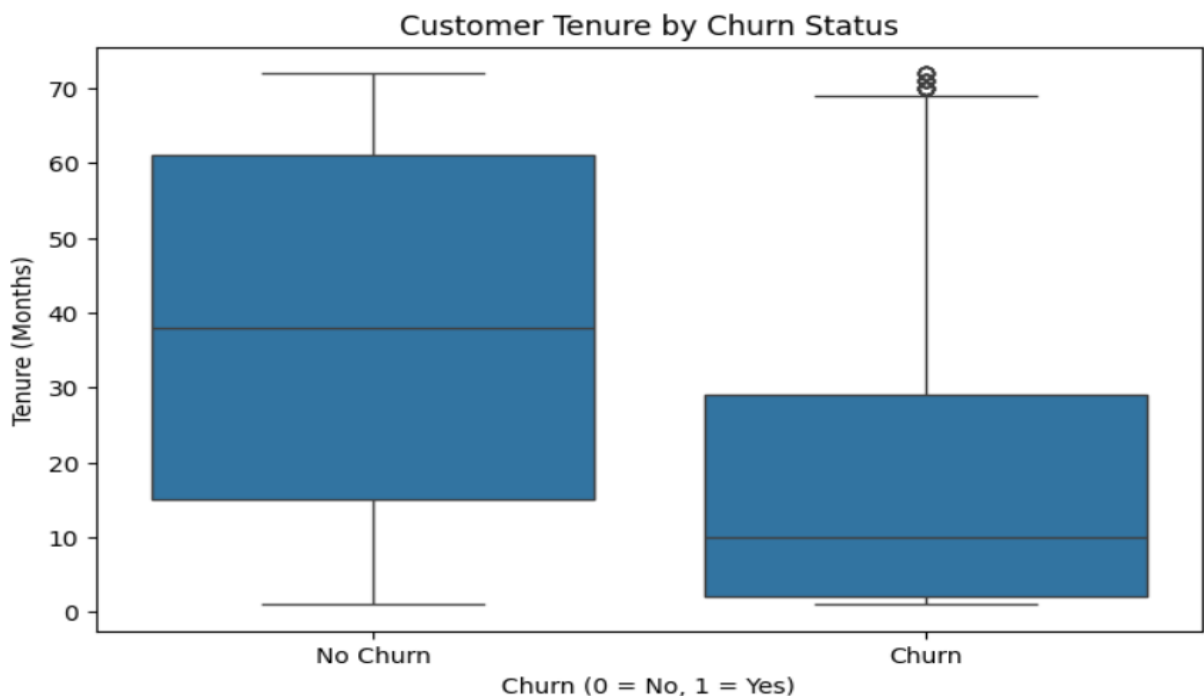


Figure 5 — Tenure vs Churn

Figure 6 — Internet Service Type vs Churn

Churn patterns varied across internet service categories. The distribution of churned and non-churned customers differed between internet service types, indicating that internet service type is an important categorical variable to consider in subsequent statistical analysis and modelling.

```
# Internet service type vs customer churn
```

```
plt.figure(figsize=(8, 5))
```

```
sns.countplot(data=df_model, x="InternetService", hue="Churn")
```

```
plt.title("Customer Churn by Internet Service Type")
```

```
plt.xlabel("Internet Service")
```

```
plt.ylabel("Number of Customers")
```

```
plt.legend(title="Churn", labels=["No", "Yes"])
```

```
plt.show()
```

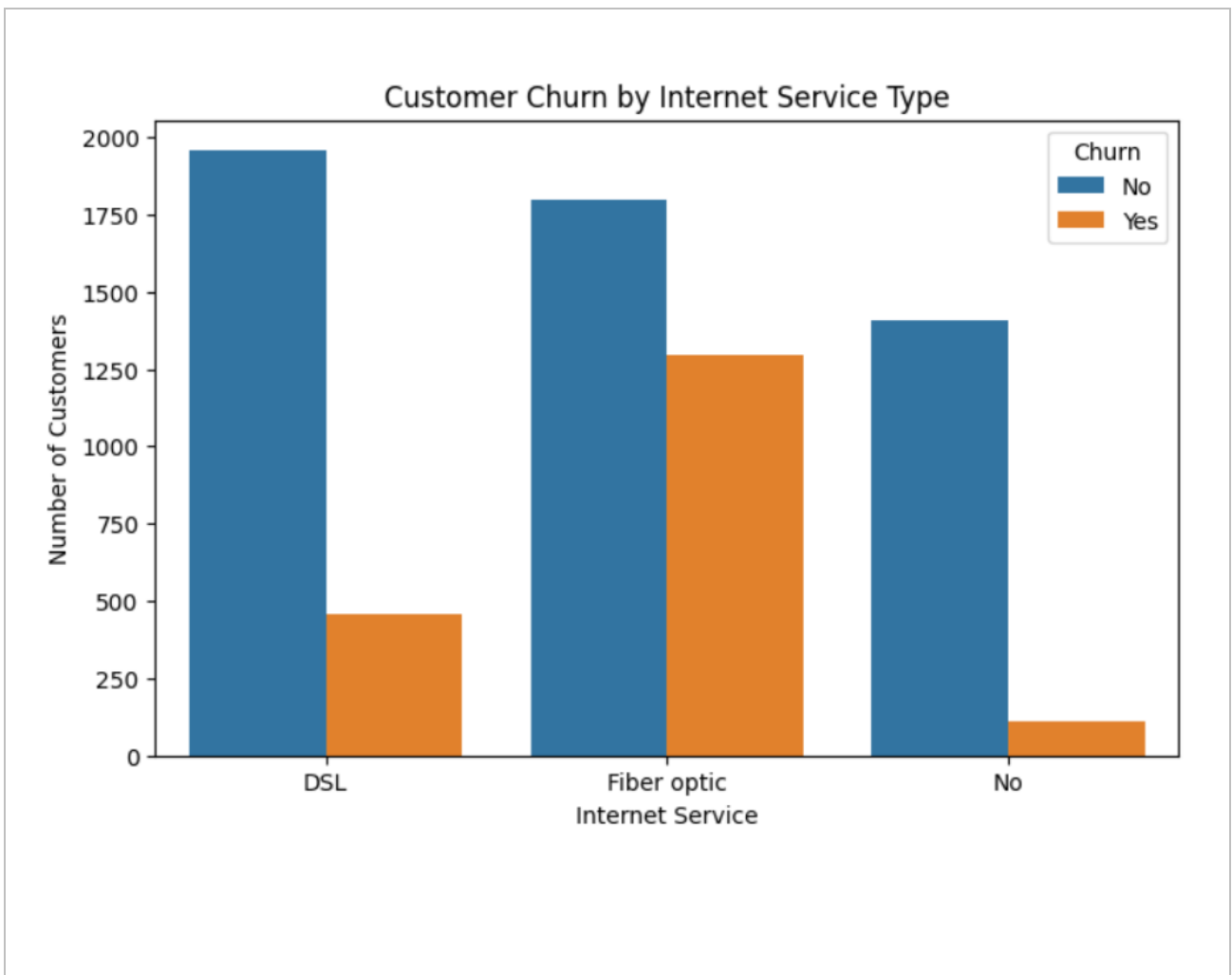
**Figure 6 — Internet Service Type vs Churn**

Figure 7 — Payment Method vs Churn

The distribution of churn also varied across payment methods. Some payment categories showed a higher proportion of churned customers than others. This provides an additional customer-behaviour pattern for investigation, although the observed association does not imply a direct causal effect.

```
# Payment method vs customer churn
```

```
plt.figure(figsize=(10, 5))
```

```
sns.countplot(data=df_model, x="PaymentMethod", hue="Churn")
```

```
plt.title("Customer Churn by Payment Method")
```

```
plt.xlabel("Payment Method")
```

```
plt.ylabel("Number of Customers")
```

```
plt.legend(title="Churn", labels=["No", "Yes"])
```

```
plt.xticks(rotation=20)
```

```
plt.show()
```

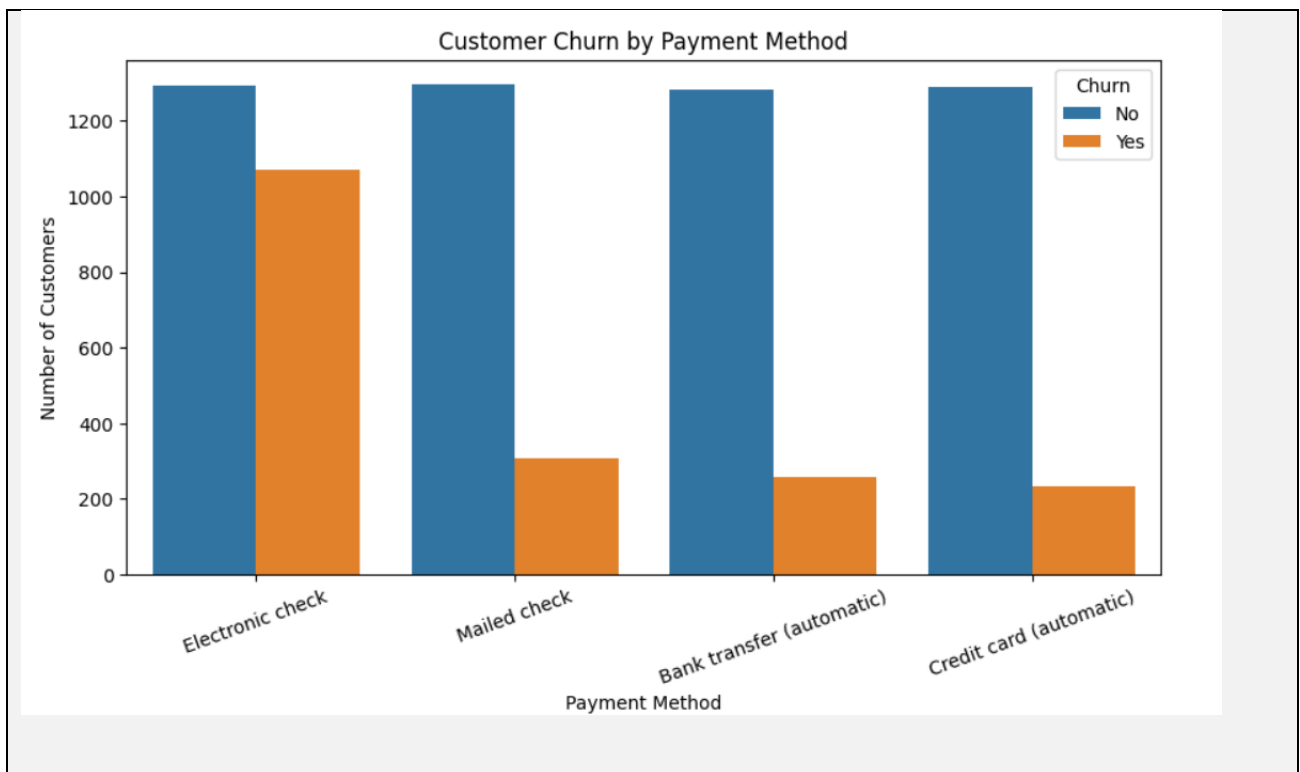


Figure 7 — Payment Method vs Churn

9. STATISTICAL ANALYSIS

Statistical analysis was conducted to summarize the characteristics of the cleaned dataset and examine differences between churned and non-churned customers. Descriptive statistics including mean and median were calculated for key numerical variables. The analysis focused on tenure, MonthlyCharges, and TotalCharges because these variables provide useful information about customer duration and billing behaviour.

The results show noticeable differences between the two churn groups. The mean tenure of non-churned customers was **37.65 months**, compared with **17.98 months** for churned customers. Similarly, the mean MonthlyCharges was **61.31** for non-churned customers and **74.44** for churned customers. The mean TotalCharges was **2555.34** for non-churned customers compared with **1531.80** for churned customers. The lower TotalCharges among churned customers is consistent with their shorter average tenure.

The median values also show similar patterns, particularly for tenure and MonthlyCharges. These descriptive differences were further examined using statistical significance tests.

Variable	Churn Status	Mean	Median
Tenure	No Churn	37.65	38.00
Tenure	Churn	17.98	10.00
MonthlyCharges	No Churn	61.31	64.45
MonthlyCharges	Churn	74.44	79.65
TotalCharges	No Churn	2555.34	1683.60
TotalCharges	Churn	1531.80	703.55

Table 4. Descriptive Statistics of Key Variables

Table 4 indicates that churned customers generally had shorter service tenure and higher monthly charges than non-churned customers. The difference between mean and median values also suggests that the distributions are not perfectly symmetric, particularly for TotalCharges. Therefore, both measures were considered when interpreting the central tendency of the variables.

9.1 Statistical Significance Testing

To determine whether the observed numerical differences were statistically significant, the Mann–Whitney U test was applied to Tenure and MonthlyCharges. The test results were statistically significant for both variables ($p < 0.001$). For Tenure, the Mann–Whitney U statistic was **2,494,979**, while for MonthlyCharges it was **5,986,148.5**.

Chi-square tests were used to examine associations between categorical variables and churn status. The association between Contract and Churn was statistically significant ($\chi^2 = 1,179.55$, $df = 2$, $p < 0.001$). Similarly, InternetService and Churn showed a statistically significant association ($\chi^2 = 728.70$, $df = 2$, $p < 0.001$).

These results indicate that the observed differences and categorical associations are unlikely to be explained solely by random variation under the assumptions of the respective tests. However, statistical significance does not establish a causal relationship between these variables and customer churn.

10. FEATURE ENGINEERING

Feature engineering was performed to create an additional variable that represents the customer's average monthly spending based on their total charges and service tenure. The engineered feature was created to provide an alternative representation of customer billing behaviour that could be used alongside the original billing variables during modelling.

Feature Name	Description	Calculation	Purpose
AvgMonthlySpend	Estimated average monthly spending of a customer	TotalCharges ÷ tenure	Represents average customer spending over the service period

Table 5. Engineered Features

The AvgMonthlySpend feature was calculated by dividing TotalCharges by tenure. For customers with zero tenure, the value was replaced with their MonthlyCharges to avoid division-by-zero and provide a meaningful value. After feature engineering, the dataset contained **20 predictor features** and the target variable Churn. The engineered feature was included in the preprocessing and modelling pipeline.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

This project is formulated as a **binary classification problem**, where the objective is to classify customers according to their churn status. The target variable is Churn, which contains two classes: Yes, indicating that the customer churned, and No, indicating that the customer did not churn.

The predictor variables consist of customer demographic information, service subscription characteristics, contract and payment information, tenure, billing variables, and the engineered AvgMonthlySpend feature. The customerID variable was excluded because it is an identifier and does not provide meaningful predictive information.

The classification formulation is appropriate because the outcome variable represents two possible customer states: churn or non-churn.

11.2 Model Selection

Two classification algorithms were selected for the analysis: **Logistic Regression** and **Decision Tree Classifier**.

Logistic Regression was selected as a baseline classification model because it is a widely used method for binary classification and provides a relatively simple approach for modelling the relationship between predictor variables and the probability of belonging to a class.

A Decision Tree Classifier was also developed to provide a non-linear modelling approach. Decision trees can represent decision rules and interactions between variables without requiring a linear relationship between the predictors and the target.

Using two different classification approaches allows their performance to be evaluated using multiple classification metrics rather than relying on a single model.

11.3 Training the Model

The models were trained using the preprocessed training dataset containing **5,625 customer records**. The preprocessing pipeline transformed the original predictor variables into 46 numerical features through categorical encoding and numerical scaling.

Logistic Regression was trained with `max_iter=1000` to allow sufficient iterations for model convergence. A Decision Tree Classifier was trained with `max_depth=5` to control the complexity of the tree and reduce the risk of excessive fitting to the training data. Both models used `random_state=42` where applicable to ensure reproducibility.

After training, predictions were generated on the unseen testing dataset containing **1,407 records**. Model performance was subsequently evaluated using accuracy, precision, recall, F1-score, and confusion matrices.

12. MODEL EVALUATION

12.1 Classification Evaluation

The classification models were evaluated on the unseen testing dataset using accuracy, precision, recall, F1-score, and confusion matrix analysis. These metrics provide different perspectives on model performance and are particularly useful when the target classes are not equally distributed.

Logistic Regression achieved a testing accuracy of **80.38%**, precision of **64.85%**, recall of **57.22%**, and F1-score of **60.80%**. Its confusion matrix contained **917 true negatives, 116 false positives, 160 false negatives, and 214 true positives**.

The Decision Tree Classifier achieved a testing accuracy of **78.39%**, precision of **58.62%**, recall of **63.64%**, and F1-score of **61.03%**.

The results show that the two models have different performance characteristics across the evaluation metrics. Therefore, accuracy alone does not provide a complete representation of their classification performance.

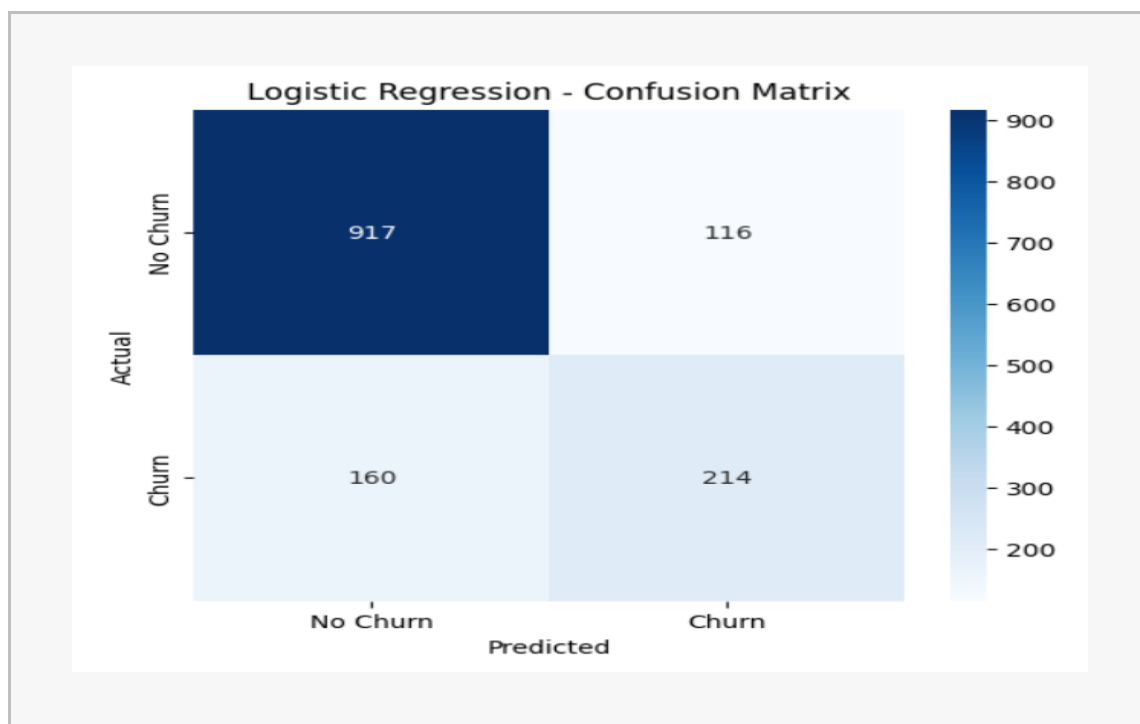


Figure 8. Confusion Matrix for Logistic Regression

The confusion matrix shows the number of correctly and incorrectly classified customers. True negatives and true positives represent correctly classified non-churned and churned customers, respectively, while false positives and false negatives represent classification errors.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	80.38%	64.85%	57.22%	60.80%
Decision Tree	78.39%	58.62%	63.64%	61.03%

Table 6. Classification Model Performance

12.2 Overfitting and Underfitting

Training and testing performance were compared to assess potential overfitting or underfitting. Logistic Regression achieved **80.41% training accuracy** and **80.38% testing accuracy**, resulting in a very small difference of approximately **0.03 percentage points**. This indicates that its performance remained highly consistent between the training and testing datasets.

The Decision Tree achieved **80.07% training accuracy** and **78.39% testing accuracy**, giving a difference of approximately **1.68 percentage points**. The larger difference suggests some reduction in performance on unseen data, although the gap remains relatively modest.

Based on the accuracy comparison, neither model shows evidence of severe overfitting. The Decision Tree shows a greater training-testing difference than Logistic Regression, while both models maintain reasonably similar performance between the two datasets.

13. RESULTS AND FINDINGS

The analysis identified several important patterns in customer churn behaviour and evaluated the ability of two classification models to predict churn status. The main findings from the exploratory, statistical, and machine learning analyses are summarized below.

1. Customer Churn Distribution

Out of the 7,032 customers in the cleaned dataset, **1,869 customers (26.58%) churned**, while **5,163 customers (73.42%) did not churn**. This indicates that the dataset contains a larger proportion of non-churned customers.

2. Tenure and Monthly Charges

Churned customers had substantially shorter service tenure than non-churned customers. The mean tenure was **17.98 months** for churned customers compared with **37.65 months** for non-churned customers. Churned customers also had higher mean monthly charges (**74.44**) than non-churned customers (**61.31**). The Mann–Whitney U tests indicated statistically significant differences for both variables ($p < 0.001$).

3. Contract Type and Internet Service

Contract type showed a statistically significant association with churn ($\chi^2 = 1,179.55$, $df = 2$, $p < 0.001$). Internet service type also showed a statistically significant association with churn ($\chi^2 = 728.70$, $df = 2$, $p < 0.001$). Exploratory analysis showed noticeable differences in churn patterns across the categories of these variables.

4. Relationship Between Numerical Variables

Correlation analysis identified a strong positive relationship between `tenure` and `TotalCharges` ($r = 0.83$). `MonthlyCharges` and `TotalCharges` also showed a positive relationship ($r = 0.65$), while the relationship between `tenure` and `MonthlyCharges` was weaker ($r = 0.25$). These relationships indicate that some numerical predictors contain overlapping information.

5. Machine Learning Performance

Both Logistic Regression and Decision Tree models were evaluated on the unseen test dataset. Logistic Regression achieved **80.38% accuracy, 64.85% precision, 57.22% recall, and 60.80% F1-score**. The Decision Tree achieved **78.39% accuracy, 58.62% precision, 63.64% recall, and 61.03% F1-score**. The results demonstrate that the models captured different aspects of the churn classification problem, with differences across the evaluation metrics.

6. Overall Finding

Overall, the analysis demonstrates that customer churn status is associated with several customer and subscription characteristics, particularly tenure, monthly charges, contract type, and internet service type. Statistical testing provided evidence for significant differences and associations in selected variables, while the machine learning models demonstrated the feasibility of classifying churn status using the available customer information.

14. DISCUSSION

The findings of this study provide an overall understanding of the factors associated with customer churn and the performance of the developed classification models. The exploratory analysis showed that churn was more common among customers with shorter service tenure and higher monthly charges. Contract type and internet service type also showed statistically significant associations with churn.

The statistical analysis supported the observed patterns. The Mann–Whitney U tests indicated statistically significant differences in tenure and MonthlyCharges between churned and non-churned customers. The chi-square tests also indicated statistically significant associations between Churn and both Contract and InternetService. These results support the presence of meaningful relationships in the dataset, although they do not establish causal relationships.

The machine learning analysis demonstrated that both Logistic Regression and Decision Tree could classify customer churn using the available customer characteristics. Logistic Regression produced 80.38% testing accuracy, while Decision Tree produced 78.39%. In contrast, the Decision Tree produced a slightly higher recall and F1-score than Logistic Regression. This shows that model performance depends on the evaluation metric considered rather than accuracy alone.

The findings also highlight relationships among numerical variables. In particular, the strong correlation between tenure and TotalCharges suggests that these variables contain overlapping information. Such relationships should be considered when interpreting model results and designing future modelling approaches.

Overall, the results demonstrate how exploratory analysis, statistical testing, feature engineering, and machine learning can be combined to investigate customer churn systematically. The analysis provides evidence that customer and subscription characteristics contain useful information for churn classification, while also highlighting the importance of evaluating models using multiple performance measures.

15. LIMITATIONS

The study has several limitations that should be considered when interpreting the findings.

- First, the analysis is based on a single Telco Customer Churn dataset, and therefore the findings may not generalize to other organizations, customer populations, or subscription industries.
- Second, the dataset contains observational customer information. The statistical relationships identified in the analysis should not be interpreted as causal relationships. For example, an association between monthly charges and churn does not establish that monthly charges directly cause customers to leave.
- Third, the machine learning analysis was based on two classification models: Logistic Regression and Decision Tree. Other algorithms and advanced modelling approaches were not evaluated in this study. Model performance may therefore differ if additional algorithms or optimization techniques are applied.
- Fourth, the target variable is not evenly distributed, with non-churned customers representing the majority of the cleaned dataset. Although multiple evaluation metrics were used, class imbalance may still affect model performance, particularly recall and precision.
- Finally, the engineered AvgMonthlySpend feature is mathematically derived from existing billing information and was not independently evaluated for incremental predictive value. Future studies could investigate additional feature engineering and model-selection techniques.

16. CONCLUSION

This study investigated customer churn using data science, statistical analysis, and machine learning techniques. The Telco Customer Churn dataset was cleaned and prepared by addressing invalid `TotalCharges` entries, converting data types, encoding categorical variables, scaling numerical variables, and separating the data into training and testing sets.

Exploratory and statistical analyses identified meaningful differences and associations related to customer churn. Churned customers generally had shorter service tenure and higher monthly charges than non-churned customers. Contract type and internet service type also showed statistically significant associations with churn. The correlation analysis further identified a strong positive relationship between tenure and `TotalCharges`.

Two classification models, Logistic Regression and Decision Tree, were developed to predict customer churn. Both models achieved reasonable classification performance on the unseen testing dataset, with Logistic Regression achieving 80.38% accuracy and Decision Tree achieving 78.39% accuracy. Their differences in precision, recall, and F1-score demonstrate the importance of evaluating classification models using multiple metrics.

Overall, the project demonstrates that customer subscription data can be systematically analysed to identify patterns associated with churn and to develop machine learning models for churn classification. The results provide a data-driven foundation for further investigation and development of customer retention strategies.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

Based on the findings of this study, the following recommendations can be made:

1. **Focus on customers with shorter tenure:** The analysis showed substantially lower average tenure among churned customers. Customer retention efforts can therefore pay particular attention to customers during the early stages of their subscription.
2. **Examine month-to-month contract customers:** Contract type showed a statistically significant association with churn. Customers on shorter-term contracts can be analysed separately to understand their retention requirements.
3. **Review customers with higher monthly charges:** Churned customers had higher average MonthlyCharges than non-churned customers. Service providers can examine the relationship between pricing, subscribed services, and customer retention.
4. **Use multiple indicators for churn analysis:** Tenure, monthly charges, contract type, and internet service type showed useful relationships with churn. Considering multiple customer characteristics together can provide a more comprehensive view of churn risk.
5. **Evaluate models using multiple metrics:** Accuracy alone does not fully describe classification performance. Precision, recall, F1-score, and confusion matrices should be considered when evaluating churn prediction models.

17.2 Future Work

Future work can extend this study by evaluating additional machine learning algorithms such as Random Forest, Support Vector Machine, and Gradient Boosting. Hyperparameter tuning and cross-validation can also be applied to obtain more robust performance estimates.

Additional customer variables, such as service usage, customer-support interactions, complaints, and historical subscription changes, could be incorporated if available. Further feature engineering and feature-selection techniques could also be investigated.

Future studies could address the class imbalance in the target variable using appropriate sampling or model-based techniques and evaluate additional metrics such as ROC-AUC and Precision-Recall AUC.

Finally, the developed approach could be tested on an independent dataset and potentially integrated into a practical churn-monitoring system. This would help evaluate how well the modelling approach generalizes to new customer data and real-world applications.

18. REFERENCES

1. International Business Machines Corporation (IBM). (n.d.). *Telco customer churn*. IBM Cognos Analytics Samples. IBM Documentation.
2. Ahmad, A. K., Jafar, A., & Aljoumaa, K. (2019). Customer churn prediction in telecom using machine learning in big data platform. *Journal of Big Data*, 6, 28.
<https://doi.org/10.1186/s40537-019-0191-6>
3. Scikit-learn developers. (2026). *LogisticRegression — scikit-learn 1.9.1 documentation*. Scikit-learn.
4. Scikit-learn developers. (2026). *DecisionTreeClassifier — scikit-learn 1.9.1 documentation*. Scikit-learn.
5. Scikit-learn developers. (2026). *Confusion matrix — scikit-learn 1.9.1 documentation*. Scikit-learn.
6. Scikit-learn developers. (2026). *OneHotEncoder — scikit-learn 1.9.1 documentation*. Scikit-learn.
7. SciPy developers. (2026). *mannwhitneyu — SciPy v1.18.0 documentation*. SciPy.
8. SciPy developers. (2026). *chi2_contingency — SciPy v1.18.0 documentation*. SciPy.
9. Mariani, G., et al. (2024). A decade of churn prediction techniques in the TelCo domain: A survey. *SN Computer Science*. <https://doi.org/10.1007/s42979-024-02722-7>
10. Zerine, I., Islam, M. M., Khan, M. A. U., et al. (2026). Explainable churn prediction in telecom with tabular ML five model benchmark and SHAP analysis. *Discover Artificial Intelligence*, 6, 263.
11. Mettle. (2026). Machine learning-based customer churn prediction in telecommunication industry. *Applied Computational Intelligence and Soft Computing*.
<https://doi.org/10.1155/acis/8496186>
12. India Data Research Academy (IDRA). (2026). *IDRA Capstone Project Guide 2026*.
13. India Data Research Academy (IDRA). (2026). *IDRA Capstone Project Report Template Sample*.

APPENDIX A. CODE

A.1 Data Loading and Inspection

```
# Import required libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    confusion_matrix,
    classification_report
)

print("Libraries imported successfully.")

# Load the Telco Customer Churn dataset

file_path = "/content/P_4_WA_Fn-UseC_-Telco-Customer-Churn.csv"

df = pd.read_csv(file_path)

print("Dataset loaded successfully!")
print("Shape:", df.shape)

# Basic dataset inspection

print("Dataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nFirst 5 Rows:")
display(df.head())

print("\nData Types:")
display(df.dtypes)

print("\nBasic Statistical Summary:")
display(df.describe(include="all").T)
```

A.2 Data Cleaning

```
# Detailed data quality check

print("==== MISSING VALUES ====")
missing = df.isnull().sum()
print(missing[missing > 0])

print("\n==== DUPLICATE RECORDS ====")
print("Number of duplicate rows:", df.duplicated().sum())

print("\n==== DATA TYPES ====")
display(df.dtypes)

print("\n==== UNIQUE VALUES ====")
for column in df.columns:
    print(f"\n{column}:")
    print(df[column].unique()[:10])

# Convert TotalCharges to numeric
# Invalid/blank values will become NaN

df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

print("Missing values in TotalCharges after conversion:")
print(df["TotalCharges"].isnull().sum())

print("\nData type of TotalCharges:")
print(df["TotalCharges"].dtype)

# Check rows affected by TotalCharges conversion

print("Missing TotalCharges after conversion:")
print(df["TotalCharges"].isnull().sum())

print("\nTotalCharges data type:")
print(df["TotalCharges"].dtype)

# Handle missing TotalCharges values

print("Rows with missing TotalCharges:")
display(
    df[
        df["TotalCharges"].isnull()
    ][
        ["customerID", "tenure", "MonthlyCharges", "TotalCharges"]
    ]
)

# Remove rows with missing TotalCharges

df = df.dropna(subset=["TotalCharges"])

print("\nDataset shape after cleaning:")
print(df.shape)

print("\nRemaining missing values:")
print(df.isnull().sum().sum())

# Final dataset validation

print("==== FINAL DATASET CHECK ====")
print("Shape:", df.shape)
```

```

print("Duplicate rows:", df.duplicated().sum())
print("Total missing values:", df.isnull().sum().sum())

print("\nFinal data types:")
display(df.dtypes)

# ===== OUTLIER CHECK USING IQR METHOD =====

numeric_columns = ["tenure", "MonthlyCharges", "TotalCharges"]

outlier_results = []

for column in numeric_columns:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[
        (df[column] < lower_bound) |
        (df[column] > upper_bound)
    ]

    outlier_results.append({
        "Variable": column,
        "Q1": Q1,
        "Q3": Q3,
        "IQR": IQR,
        "Lower Bound": lower_bound,
        "Upper Bound": upper_bound,
        "Outlier Count": len(outliers)
    })

outlier_summary = pd.DataFrame(outlier_results)

display(outlier_summary)

```

A.3 Preprocessing

```

# 9. Feature Engineering

df_model_fe = df_model.copy()

# Average monthly spending over the customer's tenure
df_model_fe["AvgMonthlySpend"] = (
    df_model_fe["TotalCharges"] /
    df_model_fe["tenure"].replace(0, np.nan)
)

# Replace undefined values for zero-tenure customers
df_model_fe["AvgMonthlySpend"] = df_model_fe["AvgMonthlySpend"].fillna(
    df_model_fe["MonthlyCharges"]
)

print("Feature engineering completed.")
print("\nNew feature: AvgMonthlySpend")

display(
    df_model_fe[
        ["tenure", "MonthlyCharges", "TotalCharges", "AvgMonthlySpend"]
    ].head()
)

```

```

# Validate engineered feature

print("Original features:", df_model.shape[1] - 1)
print("Features after engineering:", df_model_fe.shape[1] - 1)

print("\nMissing values in engineered dataset:")
print(df_model_fe.isnull().sum().sum())

print("\nEngineered dataset shape:")
print(df_model_fe.shape)

# Prepare features and target for machine learning

X = df_model_fe.drop(columns=["Churn"])
y = df_model_fe["Churn"]

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)

print("\nTraining target distribution:")
print(y_train.value_counts())

print("\nTesting target distribution:")
print(y_test.value_counts())

# Identify numerical and categorical features

numerical_features = X_train.select_dtypes(
    include=["int64", "float64"]
).columns.tolist()

categorical_features = X_train.select_dtypes(
    include=["object"]
).columns.tolist()

print("Numerical features:")
print(numerical_features)

print("\nCategorical features:")
print(categorical_features)

print("\nNumber of numerical features:", len(numerical_features))
print("Number of categorical features:", len(categorical_features))

# Create preprocessing pipeline

from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer

preprocessor = ColumnTransformer(
    transformers=[
        ("num", StandardScaler(), numerical_features),
        ("cat", OneHotEncoder(handle_unknown="ignore"), categorical_features)
    ]
)

```

```
print("Preprocessing pipeline created successfully.")
print("Numerical features will be standardized.")
print("Categorical features will be one-hot encoded.")

# Apply preprocessing to training and testing data

X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed = preprocessor.transform(X_test)

print("Preprocessing applied successfully.")
print("Processed training data shape:", X_train_processed.shape)
print("Processed testing data shape:", X_test_processed.shape)
```

A.4 EDA and Visualization

```
# Churn distribution

plt.figure(figsize=(7, 5))

sns.countplot(data=df_model, x="Churn")

plt.title("Customer Churn Distribution")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Number of Customers")

plt.show()

# Contract type vs customer churn

plt.figure(figsize=(8, 5))

sns.countplot(data=df_model, x="Contract", hue="Churn")

plt.title("Customer Churn by Contract Type")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
plt.legend(title="Churn", labels=["No", "Yes"])

plt.xticks(rotation=10)
plt.show()

# Tenure distribution by churn status

plt.figure(figsize=(8, 5))

sns.boxplot(data=df_model, x="Churn", y="tenure")

plt.title("Customer Tenure by Churn Status")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Tenure (Months)")
plt.xticks([0, 1], ["No Churn", "Churn"])

plt.show()

# Monthly charges by churn status

plt.figure(figsize=(8, 5))

sns.boxplot(data=df_model, x="Churn", y="MonthlyCharges")

plt.title("Monthly Charges by Churn Status")
plt.xlabel("Churn (0 = No, 1 = Yes)")
plt.ylabel("Monthly Charges")
plt.xticks([0, 1], ["No Churn", "Churn"])

plt.show()

# Internet service type vs customer churn

plt.figure(figsize=(8, 5))

sns.countplot(data=df_model, x="InternetService", hue="Churn")
```

```

plt.title("Customer Churn by Internet Service Type")
plt.xlabel("Internet Service")
plt.ylabel("Number of Customers")
plt.legend(title="Churn", labels=["No", "Yes"])

plt.show()

# Payment method vs customer churn

plt.figure(figsize=(10, 5))

sns.countplot(data=df_model, x="PaymentMethod", hue="Churn")

plt.title("Customer Churn by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Number of Customers")
plt.legend(title="Churn", labels=["No", "Yes"])

plt.show()

# ===== CORRELATION ANALYSIS =====

correlation_data = df[[
    "SeniorCitizen",
    "tenure",
    "MonthlyCharges",
    "TotalCharges"
]]

correlation_matrix = correlation_data.corr()

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap of Numerical Variables")
plt.tight_layout()
plt.show()

```

A.5 Model Training and Evaluation

```

# Train Logistic Regression model

from sklearn.linear_model import LogisticRegression

logistic_model = LogisticRegression(
    max_iter=1000,
    random_state=42
)

logistic_model.fit(X_train_processed, y_train)

print("Logistic Regression model trained successfully.")

# Evaluate Logistic Regression model

y_train_pred_lr = logistic_model.predict(X_train_processed)
y_test_pred_lr = logistic_model.predict(X_test_processed)

print("===== LOGISTIC REGRESSION RESULTS =====")

print("\nTraining Performance:")
print(f"Accuracy : {accuracy_score(y_train, y_train_pred_lr):.4f}")
print(f"Precision: {precision_score(y_train, y_train_pred_lr):.4f}")

```

```

print(f"Recall    : {recall_score(y_train, y_train_pred_lr):.4f}")
print(f"F1-score : {f1_score(y_train, y_train_pred_lr):.4f}")

print("\nTesting Performance:")
print(f"Accuracy : {accuracy_score(y_test, y_test_pred_lr):.4f}")
print(f"Precision: {precision_score(y_test, y_test_pred_lr):.4f}")
print(f"Recall    : {recall_score(y_test, y_test_pred_lr):.4f}")
print(f"F1-score : {f1_score(y_test, y_test_pred_lr):.4f}")

# Confusion Matrix - Logistic Regression

cm_lr = confusion_matrix(y_test, y_test_pred_lr)

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm_lr,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.title("Logistic Regression - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Train Decision Tree model

from sklearn.tree import DecisionTreeClassifier

decision_tree_model = DecisionTreeClassifier(
    random_state=42,
    max_depth=5
)

decision_tree_model.fit(X_train_processed, y_train)

print("Decision Tree model trained successfully.")

# Evaluate Decision Tree model

y_train_pred_dt = decision_tree_model.predict(X_train_processed)
y_test_pred_dt = decision_tree_model.predict(X_test_processed)

print("===== DECISION TREE RESULTS =====")

print("\nTraining Performance:")
print(f"Accuracy : {accuracy_score(y_train, y_train_pred_dt):.4f}")
print(f"Precision: {precision_score(y_train, y_train_pred_dt):.4f}")
print(f"Recall    : {recall_score(y_train, y_train_pred_dt):.4f}")
print(f"F1-score : {f1_score(y_train, y_train_pred_dt):.4f}")

print("\nTesting Performance:")
print(f"Accuracy : {accuracy_score(y_test, y_test_pred_dt):.4f}")
print(f"Precision: {precision_score(y_test, y_test_pred_dt):.4f}")
print(f"Recall    : {recall_score(y_test, y_test_pred_dt):.4f}")
print(f"F1-score : {f1_score(y_test, y_test_pred_dt):.4f}")

# Confusion Matrix - Decision Tree

cm_dt = confusion_matrix(y_test, y_test_pred_dt)

plt.figure(figsize=(6, 5))

```

```

sns.heatmap(
    cm_dt,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.title("Decision Tree - Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# Compare Logistic Regression and Decision Tree

model_comparison = pd.DataFrame({
    "Model": ["Logistic Regression", "Decision Tree"],
    "Accuracy": [
        accuracy_score(y_test, y_test_pred_lr),
        accuracy_score(y_test, y_test_pred_dt)
    ],
    "Precision": [
        precision_score(y_test, y_test_pred_lr),
        precision_score(y_test, y_test_pred_dt)
    ],
    "Recall": [
        recall_score(y_test, y_test_pred_lr),
        recall_score(y_test, y_test_pred_dt)
    ],
    "F1-Score": [
        f1_score(y_test, y_test_pred_lr),
        f1_score(y_test, y_test_pred_dt)
    ]
})

print("===== MODEL COMPARISON =====")
display(model_comparison.round(4))

# Check training vs testing performance

overfit_check = pd.DataFrame({
    "Model": ["Logistic Regression", "Decision Tree"],
    "Training Accuracy": [
        accuracy_score(y_train, y_train_pred_lr),
        accuracy_score(y_train, y_train_pred_dt)
    ],
    "Testing Accuracy": [
        accuracy_score(y_test, y_test_pred_lr),
        accuracy_score(y_test, y_test_pred_dt)
    ]
})

display(overfit_check.round(4))

```

APPENDIX B. ADDITIONAL FIGURES AND TABLES

B.1 Additional Model Evaluation Results

Figure 6. Decision Tree Confusion Matrix

Confusion Matrix - Decision Tree

```
cm_dt = confusion_matrix(y_test, y_test_pred_dt)
```

```
plt.figure(figsize=(6, 5))
```

```
sns.heatmap(
    cm_dt,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)
```

```
plt.title("Decision Tree - Confusion Matrix")
```

```
plt.xlabel("Predicted")
```

```
plt.ylabel("Actual")
```

```
plt.show()
```

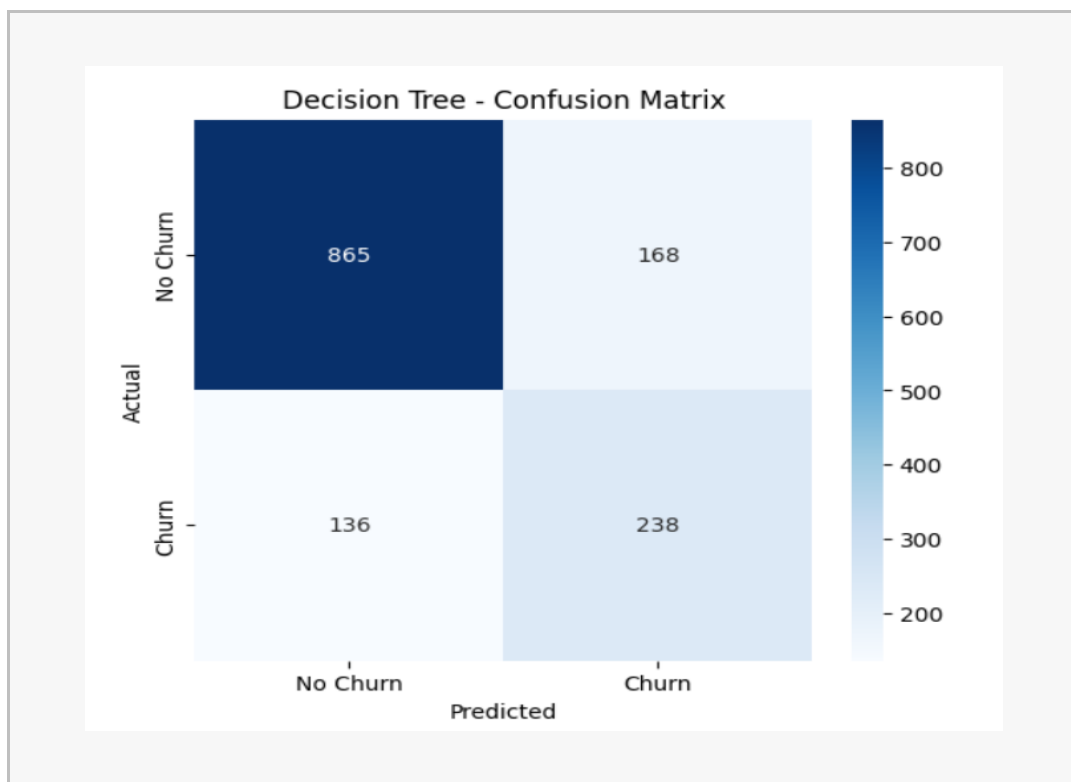


Figure 9. Decision Tree Confusion Matrix

B.2 Training and Testing Accuracy

Model	Training Accuracy	Testing Accuracy
Logistic Regression	80.41%	80.38%
Decision Tree	80.07%	78.39%

Table 7. Training and Testing Accuracy

The training and testing accuracy values provide a supplementary view of model generalization. Logistic Regression showed a very small difference between training and testing accuracy, while the Decision Tree showed a somewhat larger difference. These results support the overfitting and underfitting discussion presented in Section 12.